

Qualification Pack



Automotive IIOT Application Technician

QP Code: ASC/Q6413

Version: 3.0

NSQF Level: 3.5

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ASC/Q6413: Automotive IIOT Application Technician

Brief Job Description

Individual at this job is responsible for Integrating Machines, robots, and Automation Systems, IIOT sensors using I/O Master Link and Establish Healthy Communication using Network Protocols, Remote Monitoring and Controlling within an organization for all its processes, the new development, production and Application Phases.

Personal Attributes

The person should be organized, team-oriented and can work independently for long hours. He should be result-oriented, keen observers and have an eye for detail and quality. The individual should also be able to demonstrate skills for information order, imagination, oral expression, analytical approach, deductive reasoning, and comprehension.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [ASC/N9803: Organize work and resources \(Manufacturing\)](#)
2. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)
3. [ASC/N6432: Support in integration of machines, robots and automation system, IIOT Sensors using industrial networking protocols and I/O Link](#)
4. [ASC/N6433: Perform remote monitoring, controlling and fetching of vital machine data of the devices connected in the IIOT network](#)
5. [ASC/N6434: Carry out maintenance and troubleshooting of I/O link master and IIOT network devices](#)

Qualification Pack (QP) Parameters

Sector	Automotive
Sub-Sector	Manufacturing
Occupation	Production Engineering
Country	India
NSQF Level	3.5

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Credits	14
Aligned to NCO/ISCO/ISIC Code	NCO-2015/2144.0801
Minimum Educational Qualification & Experience	8th grade pass with 2 years of NTC with 1 Year of experience OR 8th grade pass with 3 Years of experience of relevant experience OR 10th grade pass and pursuing continuous schooling OR 11th grade pass
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	20 Years
Last Reviewed On	NA
Next Review Date	03/05/2026
NSQC Approval Date	03/05/2023
Version	3.0
Reference code on NQR	QG-3.5-AU-00535-2023-V1.1-ASDC
NQR Version	1.1

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ASC/N9803: Organize work and resources (Manufacturing)

Description

This NOS unit is about implementing safety, planning work, adopting sustainable practices for optimising use of resources

Scope

The scope covers the following :

- Maintain safe and secure working environment
- Health and hygiene
- Perform work as per quality standards
- Effective waste management practices
- Material/energy conservation practices

Elements and Performance Criteria

Maintain safe and secure working environment

To be competent, the user/individual on the job must be able to:

- PC1.** identify hazardous activities and the possible causes of risks or accidents in the workplace
- PC2.** follow safe working practices while dealing with hazards to ensure safety of self and others
- PC3.** carry out routine check of the machine for identifying potential hazards
- PC4.** use appropriate protective clothing/equipment for specific tasks and work
- PC5.** follow safety hazards and preventive techniques during fire drill
- PC6.** report any identified breaches in health, safety and security policies and procedures to the designated person

Health and hygiene

To be competent, the user/individual on the job must be able to:

- PC7.** ensure workstation and equipment are regularly clean and sanitized
- PC8.** clean hands with soap, alcohol-based sanitizer regularly
- PC9.** avoid contact with ill people and self-isolate in a similar situation
- PC10.** wear and dispose PPEs regularly and appropriately
- PC11.** report advanced hygiene and sanitation issues to appropriate authority
- PC12.** follow stress and anxiety management techniques

Perform work as per quality standards

To be competent, the user/individual on the job must be able to:

- PC13.** ensure that work is accomplished as per the requirements within the specified timeline
- PC14.** ensure team goals are given preference over individual goals

Effective waste management practices

To be competent, the user/individual on the job must be able to:

- PC15.** follow the fundamentals of 5S for waste management

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- PC16.** segregate waste into different categories
- PC17.** follow processes specified for disposal of hazardous waste
- PC18.** identify recyclable, non-recyclable and hazardous waste
- PC19.** dispose non-recyclable, recyclable and reusable waste appropriately at identified location

Material/energy conservation practices

To be competent, the user/individual on the job must be able to:

- PC20.** identify ways to optimize usage of material in various tasks/activities/processes
- PC21.** check for spills/leakages in various tasks/activities/processes
- PC22.** plug spills/leakages and escalate to appropriate authority if unable to rectify
- PC23.** check if the equipment/machine is functioning normally before commencing work and rectify wherever required
- PC24.** report malfunctioning (fumes/ sparks/emission/vibration/noise) and lapse in maintenance of equipment
- PC25.** ensure electrical equipment and appliances are properly connected and turned off when not in use

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** organisation procedures for health, safety and security, individual role and responsibilities in this context
- KU2.** the organisation's emergency procedures for different emergency situations and the importance of following the same
- KU3.** evacuation procedures for workers and visitors
- KU4.** how and when to report hazards as well as the limits of responsibility for dealing with hazards
- KU5.** potential hazards, risks and threats based on the nature of work
- KU6.** preventative and remedial actions to be taken in case of exposure to toxic material
- KU7.** various types of fire extinguisher
- KU8.** various types of safety signs and their meaning
- KU9.** appropriate first aid treatment relevant to different condition e.g. bleeding, minor burns, eye injuries etc.
- KU10.** relevant standards, procedures and policies related to 5S followed in the company
- KU11.** the various materials used and their storage norms
- KU12.** efficient utilisation of material and water
- KU13.** basics of electricity and prevalent energy efficient devices
- KU14.** common practices of conserving electricity
- KU15.** common sources and ways to minimize pollution
- KU16.** categorisation of waste into dry, wet, recyclable, non-recyclable and items of single-use plastics
- KU17.** usage of different colors of dustbins

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KU18. waste management techniques

KU19. significance of greening

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read safety instructions/guidelines

GS2. modify work practices to improve them

GS3. ask for clarifications from superior about the job requirement

GS4. work with supervisors/team members to carry out work related tasks

GS5. complete tasks efficiently and accurately within stipulated time

GS6. inform/report to concerned person in case of any problem

GS7. make timely decisions for efficient utilization of resources

GS8. write reports such as accident report, in at least English/regional language

GS9. be punctual and utilize time efficiently

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Maintain safe and secure working environment</i>	11	5	-	7
PC1. identify hazardous activities and the possible causes of risks or accidents in the workplace	2	1	-	2
PC2. follow safe working practices while dealing with hazards to ensure safety of self and others	2	-	-	1
PC3. carry out routine check of the machine for identifying potential hazards	2	1	-	1
PC4. use appropriate protective clothing/equipment for specific tasks and work	2	1	-	1
PC5. follow safety hazards and preventive techniques during fire drill	2	1	-	1
PC6. report any identified breaches in health, safety and security policies and procedures to the designated person	1	1	-	1
<i>Health and hygiene</i>	7	5	-	2
PC7. ensure workstation and equipment are regularly clean and sanitized	2	2	-	1
PC8. clean hands with soap, alcohol-based sanitizer regularly	1	1	-	1
PC9. avoid contact with ill people and self-isolate in a similar situation	1	-	-	-
PC10. wear and dispose PPEs regularly and appropriately	1	-	-	-
PC11. report advanced hygiene and sanitation issues to appropriate authority	1	1	-	-
PC12. follow stress and anxiety management techniques	1	1	-	-
<i>Perform work as per quality standards</i>	5	3	-	2
PC13. ensure that work is accomplished as per the requirements within the specified timeline	2	2	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. ensure team goals are given preference over individual goals	3	1	-	1
<i>Effective waste management practices</i>	15	10	-	4
PC15. follow the fundamentals of 5S for waste management	3	2	-	1
PC16. segregate waste into different categories	2	1	-	-
PC17. follow processes specified for disposal of hazardous waste	2	2	-	1
PC18. identify recyclable, non-recyclable and hazardous waste	4	2	-	1
PC19. dispose non-recyclable, recyclable and reusable waste appropriately at identified location	4	3	-	1
<i>Material/energy conservation practices</i>	12	7	-	5
PC20. identify ways to optimize usage of material in various tasks/activities/processes	2	1	-	1
PC21. check for spills/leakages in various tasks/activities/processes	2	1	-	1
PC22. plug spills/leakages and escalate to appropriate authority if unable to rectify	2	1	-	-
PC23. check if the equipment/machine is functioning normally before commencing work and rectify wherever required	2	2	-	1
PC24. report malfunctioning (fumes/sparks/emission/vibration/noise) and lapse in maintenance of equipment	2	1	-	1
PC25. ensure electrical equipment and appliances are properly connected and turned off when not in use	2	1	-	1
NOS Total	50	30	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	ASC/N9803
NOS Name	Organize work and resources (Manufacturing)
Sector	Automotive
Sub-Sector	Generic
Occupation	Generic
NSQF Level	3
Credits	1.5
Version	3.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

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DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

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- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

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PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

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- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

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ASC/N6432: Support in integration of machines, robots and automation system, IIOT Sensors using industrial networking protocols and I/O Link

Description

This NOS unit is about supporting and performing task related to integration of machines, robots, automation systems, IIOT Sensors using I/O link master and industrial networking protocols used in manufacturing processes to meet the specification set by an organization.

Scope

The scope covers the following :

- Install the elements in different layers of industrial network architecture and protocols
- Integrate and establish communication using I/O link master
- Ensure IIOT Network Security
- Perform post-installation activities

Elements and Performance Criteria

Install the elements in different layers of industrial network architecture and protocols

To be competent, the user/individual on the job must be able to:

- PC1.** identify the devices and systems to be connected in the IIOT network
- PC2.** analyse the installed machines, automation elements, system, and robots into different layers of network architecture like field devices, control devices, network etc.
- PC3.** interpret the network consists of network devices, automation systems and robots
- PC4.** support the engineer in installation of the suitable network protocols like MODBUS, CC-LINK, Profinet, Profibus, OPC UA, MQTT etc. as per the control system requirements
- PC5.** connect the intelligent devices and system using suitable network topology like STAR, LINE, RING as per network design document and Engineer instructions
- PC6.** evaluate the impacts of IIOT network on the environment and humans

Integrate and establish communication using I/O link master and network protocols

To be competent, the user/individual on the job must be able to:

- PC7.** install and connect the automation elements like PLC, VFD, HMI, Pneumatics, Hydraulics Elements, Industrial sensors, IIOT Sensors, control devices to I/O link master via I/O link Devices as per SOP and Engineer instructions
- PC8.** support in installation of the cable between devices as per the signaling parameters like bend radius, signal ground, terminal resistor, cable length etc.
- PC9.** establish the communication between automation system, intelligent devices, and robots by doing parameter setting like baud rate, distance, station ID and station type
- PC10.** support in setting the network parameters of automation system on the device manufacturers software as per SOP and organizational guidelines
- PC11.** turn on the power of automation devices, system in the network and look for healthy communication between them

Ensure IIOT network security

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To be competent, the user/individual on the job must be able to:

- PC12.** ensure physical security of the network contains I/O Link, I/O Link Master, IIOT Sensors, machines, robots and automation system
- PC13.** protect the network from unauthorized access or malicious internet by following organizational guidelines
- PC14.** ensure that only authorized devices should be able to connect to the network

Perform post-installation activities

To be competent, the user/individual on the job must be able to:

- PC15.** support in conducting the trials of system as per the e-plan and align it with the existing or new manufacturing process
- PC16.** handover the system to production team & support the engineer to provide them training on its use and operation as per organizational guidelines and procedures
- PC17.** prepare documents and records as a reference for future development as per Engineer instructions

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** product portfolio of organization
- KU2.** company manufacturing processes
- KU3.** Standard Operation Procedures (SOP) recommended by manufacturer for using equipment / machinery in use
- KU4.** different layers of network architecture
- KU5.** types of network protocols, topology and its significance
- KU6.** design of industrial network between devices based on protocols, topology and device parameters
- KU7.** signaling parameters required to do cable installation between devices
- KU8.** allocation of device parameters like station ID, baud rate etc. to the devices connected to the network
- KU9.** device manufacturer software for network parameter settings and device communication
- KU10.** working and integration of different elements using I/O link master to the controller
- KU11.** data types like machine, process and control data from robot and automation system in the network
- KU12.** maintenance and troubleshooting procedures like hardware, self-loop back, link test etc.
- KU13.** functioning of various network devices like routers, network switch, repeaters
- KU14.** network design concepts and Parameters to be considered
- KU15.** Automation Elements like Electro Pneumatics, Hydraulics, VFD, HMI, Servo Drives

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively at the workplace

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- GS2.** attentively listen and comprehend the information given by the process managers
- GS3.** write observations and any work-related information in English/regional language
- GS4.** recognise a workplace problem and take suitable action
- GS5.** analyse and apply the information gathered from observation, experience, reasoning or communication to act efficiently
- GS6.** complete the assigned tasks in a timely and efficient manner
- GS7.** coordinate with shop floor workers and team for installing the new systems efficiently

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Install the elements in different layers of industrial network architecture and protocols</i>	9	16	-	7
PC1. identify the devices and systems to be connected in the IIOT network	1	3	-	1
PC2. analyse the installed machines, automation elements, system, and robots into different layers of network architecture like field devices, control devices, network etc.	2	3	-	1
PC3. interpret the network consists of network devices, automation systems and robots	2	3	-	2
PC4. support the engineer in installation of the suitable network protocols like MODBUS, CC-LINK, Profinet, Profibus, OPC UA, MQTT etc. as per the control system requirements	2	3	-	1
PC5. connect the intelligent devices and system using suitable network topology like STAR, LINE, RING as per network design document and Engineer instructions	1	2	-	1
PC6. evaluate the impacts of IIOT network on the environment and humans	1	2	-	1
<i>Integrate and establish communication using I/O link master and network protocols</i>	10	18	-	5
PC7. install and connect the automation elements like PLC, VFD, HMI, Pneumatics, Hydraulics Elements, Industrial sensors, IIOT Sensors, control devices to I/O link master via I/O link Devices as per SOP and Engineer instructions	2	5	-	1
PC8. support in installation of the cable between devices as per the signaling parameters like bend radius, signal ground, terminal resistor, cable length etc.	2	3	-	1
PC9. establish the communication between automation system, intelligent devices, and robots by doing parameter setting like baud rate, distance, station ID and station type	2	4	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. support in setting the network parameters of automation system on the device manufacturers software as per SOP and organizational guidelines	2	3	-	1
PC11. turn on the power of automation devices, system in the network and look for healthy communication between them	2	3	-	1
<i>Ensure IIOT network security</i>	6	10	-	5
PC12. ensure physical security of the network contains I/O Link, I/O Link Master, IIOT Sensors, machines, robots and automation system	2	4	-	2
PC13. protect the network from unauthorized access or malicious internet by following organizational guidelines	2	3	-	2
PC14. ensure that only authorized devices should be able to connect to the network	2	3	-	1
<i>Perform post-installation activities</i>	5	6	-	3
PC15. support in conducting the trials of system as per the e-plan and align it with the existing or new manufacturing process	2	2	-	1
PC16. handover the system to production team & support the engineer to provide them training on its use and operation as per organizational guidelines and procedures	1	2	-	1
PC17. prepare documents and records as a reference for future development as per Engineer instructions	2	2	-	1
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ASC/N6432
NOS Name	Support in integration of machines, robots and automation system, IIOT Sensors using industrial networking protocols and I/O Link
Sector	Automotive
Sub-Sector	Manufacturing
Occupation	Production Engineering
NSQF Level	3.5
Credits	4
Version	2.0
Last Reviewed Date	03/05/2023
Next Review Date	03/05/2026
NSQC Clearance Date	03/05/2023

Qualification Pack

ASC/N6433: Perform remote monitoring, controlling and fetching of vital machine data of the devices connected in the IIOT network

Description

This unit is about performing remote monitoring, controlling and fetching of vital machine data of the devices connected in the IIOT network.

Scope

The scope covers the following :

- Perform IIOT network assessment
- Perform Remote Monitoring and Controlling of Machines
- Fetch Vital Machine Data

Elements and Performance Criteria

Perform IIOT network assessment

To be competent, the user/individual on the job must be able to:

- PC1.** conduct on-site surveys of the IIOT network to assess the system operation
- PC2.** monitor and record the on-field device status as per SOP
- PC3.** detect all sources of network interference available in the range of IIOT network
- PC4.** eliminate the impact of network interference by following organizational guidelines

Perform remote monitoring and controlling of machines

To be competent, the user/individual on the job must be able to:

- PC5.** monitor the real time open alarm, machine status and system logs of the IIOT network
- PC6.** manage production, quality & preventive maintenance plans remotely
- PC7.** monitor life of subsystems of IIOT network with user defined limits
- PC8.** analyse the present condition of the machines, robots and automation system (cycling, idle, setup, breakdown) by following Engineer instructions

Fetch Vital Machine Data

To be competent, the user/individual on the job must be able to:

- PC9.** fetch machine power consumption report and machine spare part life utilization report by following SOP/WI
- PC10.** identify and analyse the reason of machine idleness, machine setup and machine breakdown activity in the IIOT network
- PC11.** analyse the real time feed override, consumable request, system alarm as per SOP

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

Qualification Pack

- KU1.** organizational policies, procedures, and guidelines that relate to designing and maintaining networks
- KU2.** organizational policies and procedures for sharing data, documenting network designs and fallback mechanisms
- KU3.** who to involve while monitoring and troubleshooting the network
- KU4.** the range of standard templates and tools available and how to use them
- KU5.** connectivity protocols for device-device communications
- KU6.** wired/wireless connectivity protocols for device-device or device-gateway communications
- KU7.** the network management dashboards and applications
- KU8.** network topologies, wired and wireless technologies, fiber optics, etc.
- KU9.** updated internal and external network regulations
- KU10.** how to perform network assessments

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** follow instructions, guidelines, procedures, rules, and service level agreements
- GS2.** listen effectively and communicate information accurately
- GS3.** follow rule-based decision-making processes
- GS4.** make decisions on suitable courses
- GS5.** plan and organize the work to achieve targets and meet deadlines
- GS6.** refer anomalies to the supervisor
- GS7.** ask for clarification and advice from appropriate people
- GS8.** analyze the business impact and disseminate relevant information to others
- GS9.** apply balanced judgments to different situations
- GS10.** check the work is complete and free from errors
- GS11.** work independently and collaboratively

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Perform IIOT network assessment</i>	10	17	-	6
PC1. conduct on-site surveys of the IIOT network to assess the system operation	2	5	-	1
PC2. monitor and record the on-field device status as per SOP	2	5	-	2
PC3. detect all sources of network interference available in the range of IIOT network	3	3	-	1
PC4. eliminate the impact of network interference by following organizational guidelines	3	4	-	2
<i>Perform remote monitoring and controlling of machines</i>	10	18	-	8
PC5. monitor the real time open alarm, machine status and system logs of the IIOT network	3	5	-	2
PC6. manage production, quality & preventive maintenance plans remotely	2	4	-	2
PC7. monitor life of subsystems of IIOT network with user defined limits	3	4	-	2
PC8. analyse the present condition of the machines, robots and automation system (cycling, idle, setup, breakdown) by following Engineer instructions	2	5	-	2
<i>Fetch Vital Machine Data</i>	10	15	-	6
PC9. fetch machine power consumption report and machine spare part life utilization report by following SOP/WI	4	5	-	2
PC10. identify and analyse the reason of machine idleness, machine setup and machine breakdown activity in the IIOT network	3	5	-	2
PC11. analyse the real time feed override, consumable request, system alarm as per SOP	3	5	-	2
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ASC/N6433
NOS Name	Perform remote monitoring, controlling and fetching of vital machine data of the devices connected in the IIOT network
Sector	Automotive
Sub-Sector	Manufacturing
Occupation	Production Engineering
NSQF Level	3.5
Credits	3.5
Version	2.0
Last Reviewed Date	03/05/2023
Next Review Date	03/05/2026
NSQC Clearance Date	03/05/2023

Qualification Pack

ASC/N6434: Carry out maintenance and troubleshooting of I/O link master and IIOT network devices

Description

This unit is about performing maintenance and troubleshooting of I/O link master and IIOT network devices.

Scope

The scope covers the following :

- Perform maintenance of IIOT sensors and I/O link master
- Carry out troubleshooting activities

Elements and Performance Criteria

Perform Maintenance of IIOT Edge Devices

To be competent, the user/individual on the job must be able to:

- PC1.** analyse the machine alarms - start time, end time, duration, reason with graphical view and report
- PC2.** raise alert for machine maintenance related activities via SMS/Email as per SOP and organizational guidelines
- PC3.** analyse the MTTR & MTBF report, breakdown, OEE, machine power consumption and machine spare part life utilization report

Carry out troubleshooting activities

To be competent, the user/individual on the job must be able to:

- PC4.** perform detailed troubleshooting and analysis of IIOT networks and end devices
- PC5.** identify network blind spots
- PC6.** diagnose and resolve network configuration and connectivity issues by following SOP/WI
- PC7.** perform line test on devices connected on the IIOT network
- PC8.** perform hardware test on communication modules, I/O link, and I/O link master

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** organizational policies, procedures, and guidelines that relate to designing and maintaining networks
- KU2.** organizational policies and procedures for sharing data, documenting network designs and fallback mechanisms
- KU3.** who to involve during maintenance and troubleshooting of the network
- KU4.** range of standard templates and tools available and how to use them
- KU5.** connectivity protocols for device-device communications

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- KU6.** wired/wireless connectivity protocols for device-device or device-gateway communications
- KU7.** network management dashboards and applications
- KU8.** network topologies, wired and wireless technologies, fiber optics, etc.
- KU9.** updated internal and external network regulations
- KU10.** how to perform network assessments
- KU11.** how to diagnose and resolve network issues
- KU12.** how to identify network blind spots

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** follow instructions, guidelines, procedures, rules, and service level agreements
- GS2.** listen effectively and communicate information accurately
- GS3.** follow rule-based decision-making processes
- GS4.** make decisions on suitable courses
- GS5.** plan and organize the work to achieve targets and meet deadlines
- GS6.** refer anomalies to the supervisor
- GS7.** ask for clarification and advice from appropriate people
- GS8.** analyze the business impact and disseminate relevant information to others
- GS9.** apply balanced judgments to different situations
- GS10.** check the work is complete and free from errors
- GS11.** work independently and collaboratively

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Perform Maintenance of IIOT Edge Devices</i>	13	21	-	11
PC1. analyse the machine alarms - start time, end time, duration, reason with graphical view and report	4	6	-	4
PC2. raise alert for machine maintenance related activities via SMS/Email as per SOP and organizational guidelines	5	7	-	4
PC3. analyse the MTTR & MTBF report, breakdown, OEE, machine power consumption and machine spare part life utilization report	4	8	-	3
<i>Carry out troubleshooting activities</i>	17	29	-	9
PC4. perform detailed troubleshooting and analysis of IIOT networks and end devices	4	6	-	2
PC5. identify network blind spots	3	5	-	2
PC6. diagnose and resolve network configuration and connectivity issues by following SOP/WI	3	6	-	2
PC7. perform line test on devices connected on the IIOT network	4	6	-	1
PC8. perform hardware test on communication modules, I/O link, and I/O link master	3	6	-	2
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ASC/N6434
NOS Name	Carry out maintenance and troubleshooting of I/O link master and IIOT network devices
Sector	Automotive
Sub-Sector	Manufacturing
Occupation	Production Engineering
NSQF Level	3.5
Credits	3
Version	2.0
Last Reviewed Date	03/05/2023
Next Review Date	03/05/2026
NSQC Clearance Date	03/05/2023

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down the proportion of marks for Theory and Skills Practical for each PC.
2. The assessment for the theory part will be based on the knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for the theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

Qualification Pack

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
ASC/N9803.Organize work and resources (Manufacturing)	50	30	-	20	100	15
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	10
ASC/N6432.Support in integration of machines, robots and automation system, IIOT Sensors using industrial networking protocols and I/O Link	30	50	-	20	100	25
ASC/N6433.Perform remote monitoring, controlling and fetching of vital machine data of the devices connected in the IIOT network	30	50	-	20	100	25
ASC/N6434.Carry out maintenance and troubleshooting of I/O link master and IIOT network devices	30	50	-	20	100	25
Total	160	210	-	80	450	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
PPE	Personal Protective Equipment
PwD	Person with Disability
SOP	Standard Operating Practices
PwD	Persons with Disability
I/O	Input/Output
IIOT	Industrial Internet of Things
OPC UA	OPC Unified Architecture
MQTT	MQ Telemetry Transport
PLC	Programmable Logic Controller
VFD	Variable Frequency Drive
HMI	Human-Machine Interface
MTTR	Mean Time to Recovery or Mean Time to Restore
MTBF	Mean Time Between Failures
PPE	Personal Protective Equipment
PwD	Person with Disability
SOP	Standard Operating Practices
I/O	Input/Output
IIOT	Industrial Internet of Things
OPC UA	OPC Unified Architecture
MQTT	MQ Telemetry Transport
PLC	Programmable Logic Controller

Qualification Pack

VFD	Variable Frequency Drive
HMI	Human-Machine Interface
MTTR	Mean Time to Recovery or Mean Time to Restore
MTBF	Mean Time Between Failures

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.